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How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

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Chapter 1: Circle Geometry

Lesson	Title (MINESEC progression)
1	Definitions; equation of a circle with centre at the origin
2	Equation of a circle with centre other than the origin
3	Equation of a circle given the centre and a point on the circle
4	The general equation of a circle
5	Equation of a circle on a given diameter
6	Equation of a circle passing through three points
7	Equation of a tangent to a circle
8	Conditions for a line to be a tangent to a circle
9	Parametric equations of a circle
10	Converting the parametric form to Cartesian form
11	Length of a tangent from an external point
12	Points of intersection of a circle and a line
13	Circle through the intersection of two circles and a given point
14	Points of intersection of two circles
15	Conditions for circles to touch
16	Circle through the intersection of a line and a circle
17	Locus of a point
18	The radical axis of two circles
19	Equation of a circle that touches the coordinate axes
20	Circle through two points with its centre on a given line
21	Circle through two points and touching a straight line
22	Equation of a circle touching a straight line

Lesson 1: Definitions; equation of a circle with centre at the origin

LEARNING OBJECTIVES

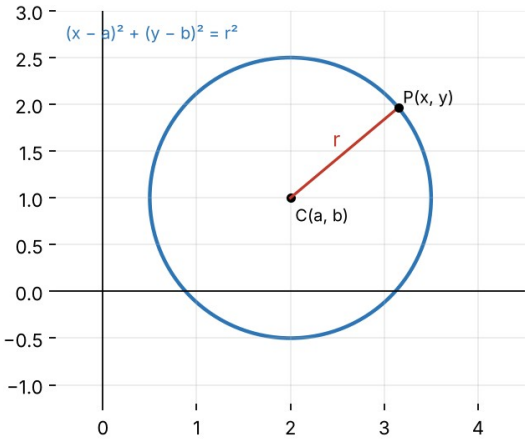
By the end of this lesson you should be able to:

- ✓ define a circle and its parts
- ✓ write the equation of a circle with centre at the origin
- ✓ decide whether a point lies inside, on or outside a circle

Lesson notes

A **circle** is the locus of points in a plane that are a fixed distance (the **radius** r) from a fixed point (the **centre**).

- A **chord** joins two points on the circle; a **diameter** is a chord through the centre; a **tangent** touches the circle at exactly one point and is perpendicular to the radius there.



KEY POINTS

Circle with centre O(0, 0) and radius r: $x^2 + y^2 = r^2$

Worked examples

Example: Write down the equation of the circle with centre O and radius 5. Does (3, −4) lie on it?

$x^2 + y^2 = 25$; 9 + 16 = 25, so yes

Example: Find the radius of $x^2 + y^2 = 18$, and say whether (3, 4) is inside, on or outside the circle.

$r = \sqrt{18} = 3\sqrt{2} \approx 4.24$; 9 + 16 = 25 > 18, so (3, 4) is outside

Practice Questions: Lesson 1

1. Write the equation of the circle with centre O and radius 7.
2. Find the radius of $x^2 + y^2 = 50$ and decide whether (5, 6) is inside, on or outside it.

Answers: *1. $x^2 + y^2 = 49$ 2. $5\sqrt{2}$; outside (25 + 36 = 61 > 50)*

Lesson 2: Equation of a circle with centre other than the origin

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write the equation of a circle with any centre and radius
- ✔ state the centre and radius from the equation

Lesson notes

KEY POINTS

Centre (a, b), radius r: $(x - a)^2 + (y - b)^2 = r^2$

Worked examples

Example: Find the equation of the circle with centre (2, −3) and radius 4.

$(x - 2)^2 + (y + 3)^2 = 16$

Example: State the centre and radius of $(x + 1)^2 + (y - 5)^2 = 7$.

Centre (−1, 5), radius $\sqrt{7}$

Practice Questions: Lesson 2

1. Find the equation of the circle with centre (−3, 4) and radius 6.
2. State the centre and radius of $(x - 5)^2 + (y + 2)^2 = 20$.

Answers: *1. $(x + 3)^2 + (y - 4)^2 = 36$ 2. (5, −2); $2\sqrt{5}$*

Lesson 3: Equation of a circle given the centre and a point on the circle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the equation of a circle given its centre and a point on it

Lesson notes

The radius is the distance from the centre to the point.

Worked examples

Example: Find the equation of the circle with centre (1, 2) passing through (4, 6).

$r^2 = 3^2 + 4^2 = 25$

$(x - 1)^2 + (y - 2)^2 = 25$

Practice Questions: Lesson 3

1. Find the equation of the circle with centre (−2, 1) passing through (1, 5).
2. Find the equation of the circle with centre (0, 3) passing through the origin.

Answers: *1. $(x + 2)^2 + (y - 1)^2 = 25$ 2. $x^2 + (y - 3)^2 = 9$*

Lesson 4: The general equation of a circle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the centre and radius from the general equation
- ✔ state the conditions for a second-degree equation to represent a circle

Lesson notes

Expanding $(x - a)^2 + (y - b)^2 = r^2$ gives the general form.

KEY POINTS

$x^2 + y^2 + 2gx + 2fy + c = 0$ is a circle with centre $(-g, -f)$ and radius $\sqrt{(g^2 + f^2 - c)}$ (needs $g^2 + f^2 - c > 0$)

A second-degree equation represents a circle only if the coefficients of x^2 and y^2 are EQUAL and there is NO xy term.

Worked examples

Example: Find the centre and radius of $x^2 + y^2 - 6x + 4y - 12 = 0$.

$2g = -6$, $g = -3$; $2f = 4$, $f = 2$; $c = -12$

Centre $(3, -2)$; $r = \sqrt{(9 + 4 + 12)} = 5$

Example: Find the centre and radius of $2x^2 + 2y^2 + 8x - 4y - 8 = 0$.

Divide by 2: $x^2 + y^2 + 4x - 2y - 4 = 0$

Centre $(-2, 1)$; $r = \sqrt{(4 + 1 + 4)} = 3$

Example: Which of these is a circle? (a) $x^2 + 2y^2 + 4x = 1$ (b) $x^2 + y^2 + xy = 4$ (c) $x^2 + y^2 + 2x + 2y + 5 = 0$

(a) No (unequal x^2, y^2 coefficients). (b) No (xy term).

(c) $g^2 + f^2 - c = 1 + 1 - 5 = -3 < 0$: no real circle

Practice Questions: Lesson 4

1. Find the centre and radius of $x^2 + y^2 + 8x - 2y + 8 = 0$.
2. Find the centre and radius of $3x^2 + 3y^2 - 6x + 12y = 0$.
3. Explain why $x^2 + y^2 - 2x + 4y + 10 = 0$ is not a real circle.

Answers: *1. $(-4, 1)$; 3 2. $(1, -2)$; $\sqrt{5}$ 3. $g^2 + f^2 - c = 1 + 4 - 10 < 0$*

Lesson 5: Equation of a circle on a given diameter

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the equation of a circle given the ends of a diameter

Lesson notes

KEY POINTS

If A(x_1, y_1) and B(x_2, y_2) are the ends of a diameter: $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

(Reason: any point P on the circle makes a right angle APB, so $\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$.)

Worked examples

Example: Find the equation of the circle with diameter AB, where A(1, 2) and B(5, −4).

$(x - 1)(x - 5) + (y - 2)(y + 4) = 0$

$x^2 - 6x + 5 + y^2 + 2y - 8 = 0$, i.e. $x^2 + y^2 - 6x + 2y - 3 = 0$

Check: centre (3, −1) = midpoint of AB ✓

Practice Questions: Lesson 5

1. Find the equation of the circle with diameter joining (−1, 3) and (5, −5).
2. Find the centre and radius of the circle in question 1.

Answers: *1. $x^2 + y^2 - 4x + 2y - 20 = 0$ 2. (2, −1); 5*

Lesson 6: Equation of a circle passing through three points

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the circle through three given points

Lesson notes

Substitute each point into $x^2 + y^2 + 2gx + 2fy + c = 0$ and solve the three equations for g, f, c.

Worked examples

Example: Find the equation of the circle through (0, 0), (4, 0) and (0, 6).

(0, 0): $c = 0$

(4, 0): $16 + 8g = 0$, $g = -2$

(0, 6): $36 + 12f = 0$, $f = -3$

$x^2 + y^2 - 4x - 6y = 0$; centre (2, 3), radius $\sqrt{13}$

Example: Find the circle through A(1, 1), B(2, −1) and C(3, 2).

A: $2 + 2g + 2f + c = 0$

B: $5 + 4g - 2f + c = 0$

C: $13 + 6g + 4f + c = 0$

B − A: $3 + 2g - 4f = 0$; C − A: $11 + 4g + 2f = 0$

Solving: $g = -5/2$, $f = -1/2$; $c = -2 + 5 + 1 = 4$

$x^2 + y^2 - 5x - y + 4 = 0$

Practice Questions: Lesson 6

1. Find the circle through (0, 0), (6, 0) and (0, 8).
2. Find the circle through (1, 0), (0, 1) and (−1, 0).

Answers: *1. $x^2 + y^2 - 6x - 8y = 0$ 2. $x^2 + y^2 = 1$*

Lesson 7: Equation of a tangent to a circle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the equation of the tangent at a point on a circle

Lesson notes

KEY POINTS

The tangent at P is perpendicular to the radius CP.

For $x^2 + y^2 = r^2$, the tangent at (x_1, y_1) is $xx_1 + yy_1 = r^2$

For $x^2 + y^2 + 2gx + 2fy + c = 0$, the tangent at (x_1, y_1) is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Worked examples

Example: Find the equation of the tangent to $x^2 + y^2 = 25$ at (3, 4).

$3x + 4y = 25$

Example: Find the tangent to $x^2 + y^2 - 4x + 2y - 20 = 0$ at (5, 3).

Centre C(2, −1); gradient of CP = (3 + 1)/(5 − 2) = 4/3

Tangent gradient = −3/4: $y - 3 = -(3/4)(x - 5)$, i.e. $3x + 4y - 27 = 0$

Practice Questions: Lesson 7

1. Find the tangent to $x^2 + y^2 = 10$ at (1, 3).
2. Find the tangent to $x^2 + y^2 + 2x - 4y - 20 = 0$ at (2, 6).

Answers: *1. $x + 3y = 10$ 2. $3x + 4y - 30 = 0$*

Lesson 8: Conditions for a line to be a tangent to a circle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ test whether a line is a tangent to a circle
- ✔ find unknown constants for tangency

Lesson notes

KEY POINTS

A line is a tangent ⇔ the perpendicular distance from the centre to the line equals the radius.

Equivalently, substituting the line into the circle gives a quadratic with discriminant 0 ($\Delta = 0$: tangent; $\Delta > 0$: cuts in two points; $\Delta < 0$: misses).

Worked examples

Example: Show that the line $3x + 4y = 15$ is a tangent to $x^2 + y^2 = 9$.

Distance from O: $|0 + 0 - 15|/\sqrt{(9 + 16)} = 15/5 = 3 = r$. Tangent ✓

Example: Find c such that $y = 2x + c$ is a tangent to $x^2 + y^2 = 5$.

$x^2 + (2x + c)^2 = 5$, $5x^2 + 4cx + c^2 - 5 = 0$

$\Delta = 16c^2 - 20(c^2 - 5) = 0$, $-4c^2 + 100 = 0$, $c = \pm 5$

Practice Questions: Lesson 8

1. Show that $5x + 12y = 26$ is a tangent to $x^2 + y^2 = 4$.
2. Find the values of c for which $y = x + c$ is a tangent to $x^2 + y^2 = 8$.

Answers: *1. distance from O = 26/13 = 2 = radius 2. c = ±4*

Lesson 9: Parametric equations of a circle

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write the parametric equations of a circle
- ✔ find points on a circle from the parameter

Lesson notes

KEY POINTS

$x^2 + y^2 = r^2$: $x = r\cos\theta$, $y = r\sin\theta$

$(x - a)^2 + (y - b)^2 = r^2$: $x = a + r\cos\theta$, $y = b + r\sin\theta$ $(0 \le \theta < 2\pi)$

Worked examples

Example: Write parametric equations for the circle with centre (3, −1) and radius 2.

$x = 3 + 2\cos\theta$, $y = -1 + 2\sin\theta$

Practice Questions: Lesson 9

1. Write parametric equations for the circle $(x + 2)^2 + (y - 1)^2 = 9$.
2. Find the point on $x = 3\cos\theta$, $y = 3\sin\theta$ where $\theta = \pi/3$.

Answers: *1. $x = -2 + 3\cos\theta$, $y = 1 + 3\sin\theta$ 2. $(3/2, 3\sqrt{3}/2)$*

Lesson 10: Converting the parametric form to Cartesian form

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ convert parametric equations of a circle to Cartesian form

Lesson notes

Use $\cos^2\theta + \sin^2\theta = 1$ to eliminate θ.

Worked examples

Example: Find the Cartesian equation of $x = 4 + 5\cos\theta$, $y = -2 + 5\sin\theta$.

$\cos\theta = (x - 4)/5$, $\sin\theta = (y + 2)/5$

$(x - 4)^2 + (y + 2)^2 = 25$

Example: Find the Cartesian equation of $x = 1 + 3\sin t$, $y = 2 - 3\cos t$.

$(x - 1)^2 + (y - 2)^2 = 9\sin^2 t + 9\cos^2 t = 9$

Practice Questions: Lesson 10

1. Find the Cartesian equation of $x = 2 + 4\cos\theta$, $y = 3 + 4\sin\theta$.
2. Find the Cartesian equation of $x = 5\sin t - 1$, $y = 5\cos t$.

Answers: *1. $(x - 2)^2 + (y - 3)^2 = 16$ 2. $(x + 1)^2 + y^2 = 25$*

Lesson 11: Length of a tangent from an external point

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the length of a tangent from an external point to a circle

Lesson notes

KEY POINTS

If $P(x_1, y_1)$ is outside the circle with centre C and radius r, the tangent length PT satisfies $PT^2 = PC^2 - r^2$

For $x^2 + y^2 + 2gx + 2fy + c = 0$: $PT^2 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ (substitute P into the left-hand side)

Worked examples

Example: Find the length of the tangent from (5, 7) to $x^2 + y^2 - 4x - 2y - 4 = 0$.

$PT^2 = 25 + 49 - 20 - 14 - 4 = 36$, $PT = 6$

Practice Questions: Lesson 11

1. Find the length of the tangent from (6, 8) to $x^2 + y^2 = 36$.
2. Find the length of the tangent from (4, −1) to $x^2 + y^2 - 2x + 4y - 4 = 0$.

Answers: *1. 8 2. 1*

Lesson 12: Points of intersection of a circle and a line

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the points where a line meets a circle
- ✔ find the length of a chord

Lesson notes

Substitute the line into the circle and solve the resulting quadratic.

Worked examples

Example: Find where the line $y = x + 1$ meets $x^2 + y^2 = 13$.

$x^2 + (x + 1)^2 = 13$, $2x^2 + 2x - 12 = 0$, $x^2 + x - 6 = 0$, $(x + 3)(x - 2) = 0$

$x = 2$, $y = 3$ and $x = -3$, $y = -2$: points $(2, 3)$ and $(-3, -2)$

Example: Find the length of the chord cut from $x^2 + y^2 = 25$ by the line $x + y = 7$.

$x^2 + (7 - x)^2 = 25$, $2x^2 - 14x + 24 = 0$, $x^2 - 7x + 12 = 0$, $x = 3$ or 4

Points $(3, 4)$ and $(4, 3)$; chord length $\sqrt{(1 + 1)} = \sqrt{2}$

Practice Questions: Lesson 12

1. Find where $y = 2x$ meets $x^2 + y^2 = 20$.
2. Find the length of the chord cut from $x^2 + y^2 = 10$ by the line $y = 1$.

Answers: *1. $(2, 4)$ and $(-2, -4)$ 2. 6*

Lesson 13: Circle through the intersection of two circles and a given point

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the circle through the intersection of two circles and a given point

Lesson notes

KEY POINTS

If $S_1 = 0$ and $S_2 = 0$ are two circles, then $S_1 + \lambda S_2 = 0$ ($\lambda \neq -1$) is a circle through their points of intersection. Choose λ using the extra condition.

Worked examples

Example: Find the circle through the intersection of $x^2 + y^2 = 4$ and $x^2 + y^2 - 2x - 4y = 0$ that passes through $(1, -1)$.

$(x^2 + y^2 - 4) + \lambda(x^2 + y^2 - 2x - 4y) = 0$

At $(1, -1)$: $(2 - 4) + \lambda(2 - 2 + 4) = 0$, $-2 + 4\lambda = 0$, $\lambda = 1/2$

Multiply by 2: $2(x^2 + y^2 - 4) + (x^2 + y^2 - 2x - 4y) = 0$

$3x^2 + 3y^2 - 2x - 4y - 8 = 0$

Practice Questions: Lesson 13

1. Find the circle through the intersection of $x^2 + y^2 = 9$ and $x^2 + y^2 + 2x - 4y = 0$ that passes through $(0, 1)$.

Answers: *1. $5x^2 + 5y^2 + 16x - 32y + 27 = 0$ ($\lambda = -8/3$)*

Lesson 14: Points of intersection of two circles

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ find the points of intersection of two circles

Lesson notes

Subtract the equations to remove x^2 and y^2: this gives a straight line (the common chord). Solve it with one of the circles.

Worked examples

Example: Find the points of intersection of $x^2 + y^2 = 10$ and $x^2 + y^2 - 4x - 2y = 0$.

Subtract: $4x + 2y = 10$, so $y = 5 - 2x$

$x^2 + (5 - 2x)^2 = 10$, $5x^2 - 20x + 15 = 0$, $x^2 - 4x + 3 = 0$

$x = 1$, $y = 3$; $x = 3$, $y = -1$. Points $(1, 3)$ and $(3, -1)$

Practice Questions: Lesson 14

1. Find the points of intersection of $x^2 + y^2 = 5$ and $x^2 + y^2 - 6x - 2y + 5 = 0$.

Answers: *1. (1, 2) and (2, −1)*

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Upper Sixth Pure Maths with Mechanics**. The complete book has every lesson of the MINESEC progression sheet, in order:

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